Title:

Multi-Layer Behavioral Fraud Detection System for Indian Digital Payments Using Machine Learning

Submitted By

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1. Introduction

Digital payment fraud in India has become a critical threat, with over ₹11,333 crore lost annually across UPI and card transactions. Despite mandatory OTP verification enforced by RBI, fraud continues to rise — primarily because 47% of cases involve social engineering where victims willingly share their OTP. Existing systems verify identity but do not analyze behavioral patterns. This project aims to develop an AI-powered system that adds a behavioral intelligence layer on top of OTP, detecting fraud in real-time by analyzing transaction patterns specific to Indian payment behavior.

2. Problem Statement

Current Indian digital payment systems rely solely on OTP as a security mechanism, which is ineffective against social engineering attacks such as vishing, SIM swap fraud, and fake UPI collect requests. There is a need for an automated behavioral analysis system that can detect suspicious transactions in real-time even when OTP authentication is successfully completed by a fraudster.

3. Proposed Approach

The system will use publicly available credit card fraud datasets combined with synthetically generated Indian transaction data. Transactions will be analyzed using behavioral features such as amount anomaly, transaction time, device fingerprint, location deviation, and spending velocity. Multiple machine learning models will be trained and compared, with the best performing model deployed as a real-time web application simulating UPI and card payment scenarios.

4. Objectives

- To study Indian digital payment fraud petterns including vishing, SIM swap, and fake UPI collection request attacks
- To preprocess and engineer behavioral features from transaction data for effective fraud pattern extraction
- To design and implement machine learning models for real-time binary fraud classification
- To handle severe class imbalance in fraud datasets using SMOTE oversampling technique
- To develop a full-stack web application simulating Indian payment scenarios with real-time fraud blocking and explainable AI using SHAP

5. Tools & Technologies

- Python
- Machine Learning Libraries (Scikit-learn, XGBoost, LightGBM)
- Deep Learning Framework (TensorFlow/Keras)
- Imbalanced-learn (SMOTE)
- SHAP (Explainability)
- Flask (Backend API)
- HTML, CSS, JavaScript (Frontend)
- SQLite (Database)
- Google Colab / Jupyter Notebook

6. Expected Outcome

The proposed system is expected to detect fraudulent UPI and card transactions in real-time with over 95% recall, block suspicious payments before processing, provide explainable fraud reasons to users, and serve as a behavioral Layer 2 security mechanism for Indian digital payment infrastructure.